

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	05 July 2026
Team ID	
Project Name	EV Analytics & Visualization Dashboard
Maximum Marks	4 Marks

Technical Architecture:

The diagram below shows the architecture of the EV Analytics & Visualization Dashboard, from data sources through cloud processing to the Tableau dashboards used by planners, fleet managers, and consumers.

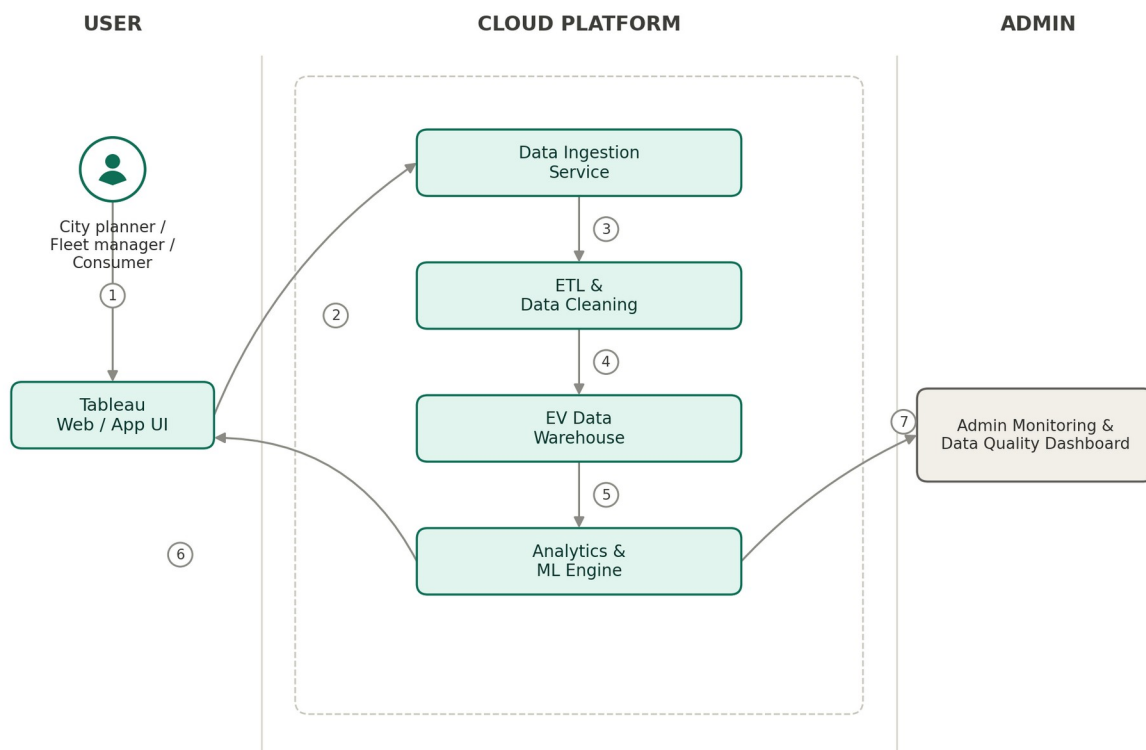


Table-1: Components & Technologies:

S. No	Component	Description	Technology
1.	User Interface	Interactive dashboards used by city planners, fleet managers, and consumers to explore EV data	Tableau Desktop / Tableau Server, HTML, CSS, JavaScript
2.	Application Logic-1	Ingests raw data from charging stations, vehicle sensors, and EV model specs	Python (Pandas, Requests), REST APIs
3.	Application Logic-2	Cleans, transforms, and aggregates raw data (ETL)	Python, SQL, Tableau Prep Builder

S. No	Component	Description	Technology
4.	Application Logic-3	Runs analytics and predictive models for battery health and range prediction	Python (scikit-learn), R
5.	Database	Structured storage of charging, battery, and vehicle data	MySQL / PostgreSQL
6.	Cloud Database	Cloud-hosted EV data warehouse for analytics at scale	AWS RDS, Google BigQuery, or Snowflake
7.	File Storage	Storage for raw sensor logs and exported reports	AWS S3 / Google Cloud Storage
8.	External API-1	Charging station location and availability data	Open Charge Map API
9.	External API-2	EV model specifications and pricing data	Manufacturer / EV database API (e.g., NREL)
10.	Machine Learning Model	Predicts battery degradation and driving range	Regression / Random Forest model (scikit-learn)
11.	Infrastructure (Server / Cloud)	Application deployment. Local Server Configuration: N/A Cloud Server Configuration: Cloud VM for ETL jobs + Tableau Server	AWS EC2, Tableau Server on Cloud

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Data processing and modeling built on open-source Python libraries	Pandas, scikit-learn, Tableau Prep Builder
2.	Security Implementations	Data encrypted at rest and in transit; access to dashboards and data restricted by role	AES-256 encryption, TLS, Tableau Server role-based access control (RBAC), OAuth2
3.	Scalable Architecture	Layered pipeline (ingestion, storage, analytics, visualization) hosted on the cloud, allowing each layer to scale independently as vehicles and users grow	Cloud data warehouse, containerized ETL jobs
4.	Availability	Cloud database replication and Tableau Server clustering keep dashboards accessible with minimal downtime	Load-balanced Tableau Server, cloud DB replicas
5.	Performance	Tableau extracts and cached queries reduce load time; scheduled ETL jobs keep the warehouse fresh without slowing dashboards	Tableau data extracts, indexed SQL queries